Computer Networks

Lecture 27 Network Layer

Dr. Tauheed Ahmed

Content

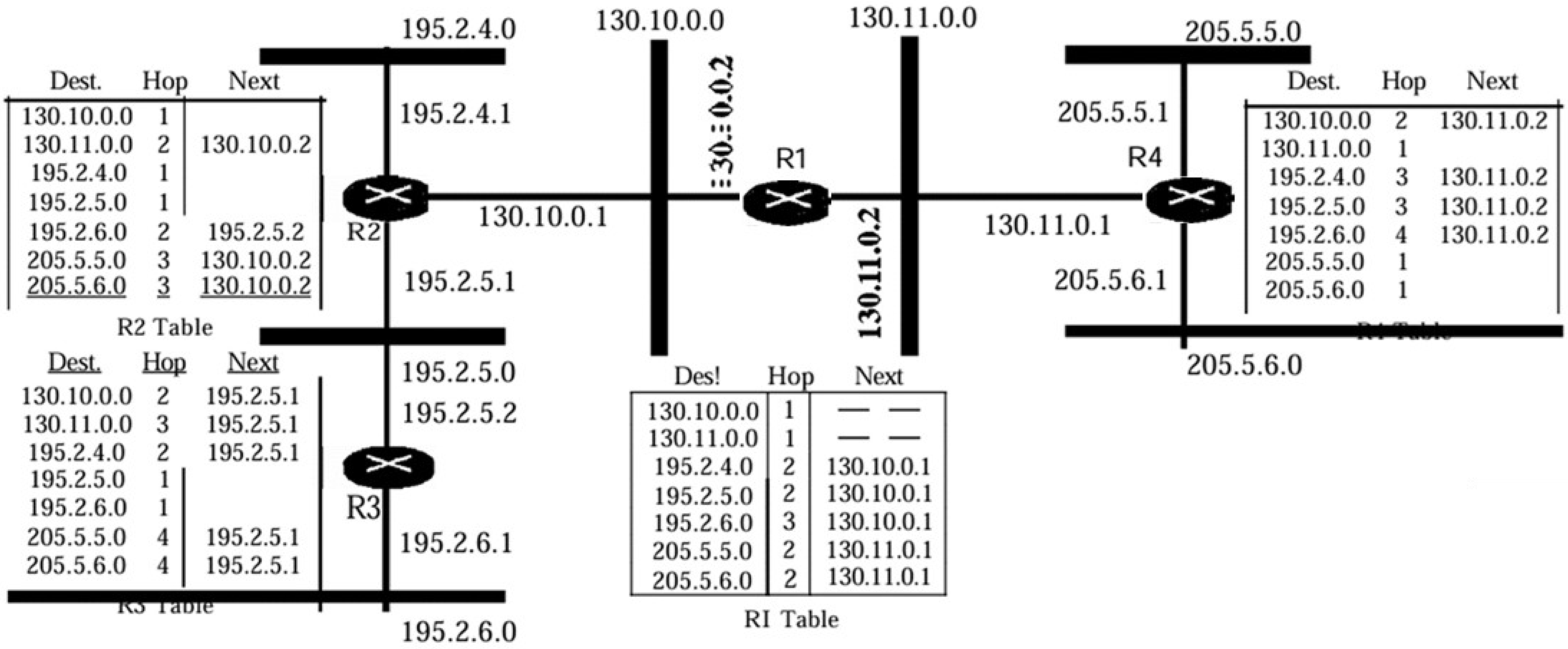
- Distance Vector Routing
  - Routing Information Protocol
- Link State Routing
  - Open Shortest Path First

Routing Information Protocol (RIP)

RIP implements distance vector routing directly with some considerations:

- In an autonomous system, we are dealing with routers and networks (links). The routers have routing tables; networks do not.
- The destination in a routing table is a network, which means the first column defines a network address.
- The metric used by RIP is very simple; the distance is defined as the number of links (networks) to reach the destination. For this reason, the metric in RIP is called a hop count.
- Infinity is defined as 16, which means that any route in an autonomous system using RIP cannot have more than 15 hops.
- The next-node column defines the address of the router to which the packet is to be sent to reach its destination.

RIP Example

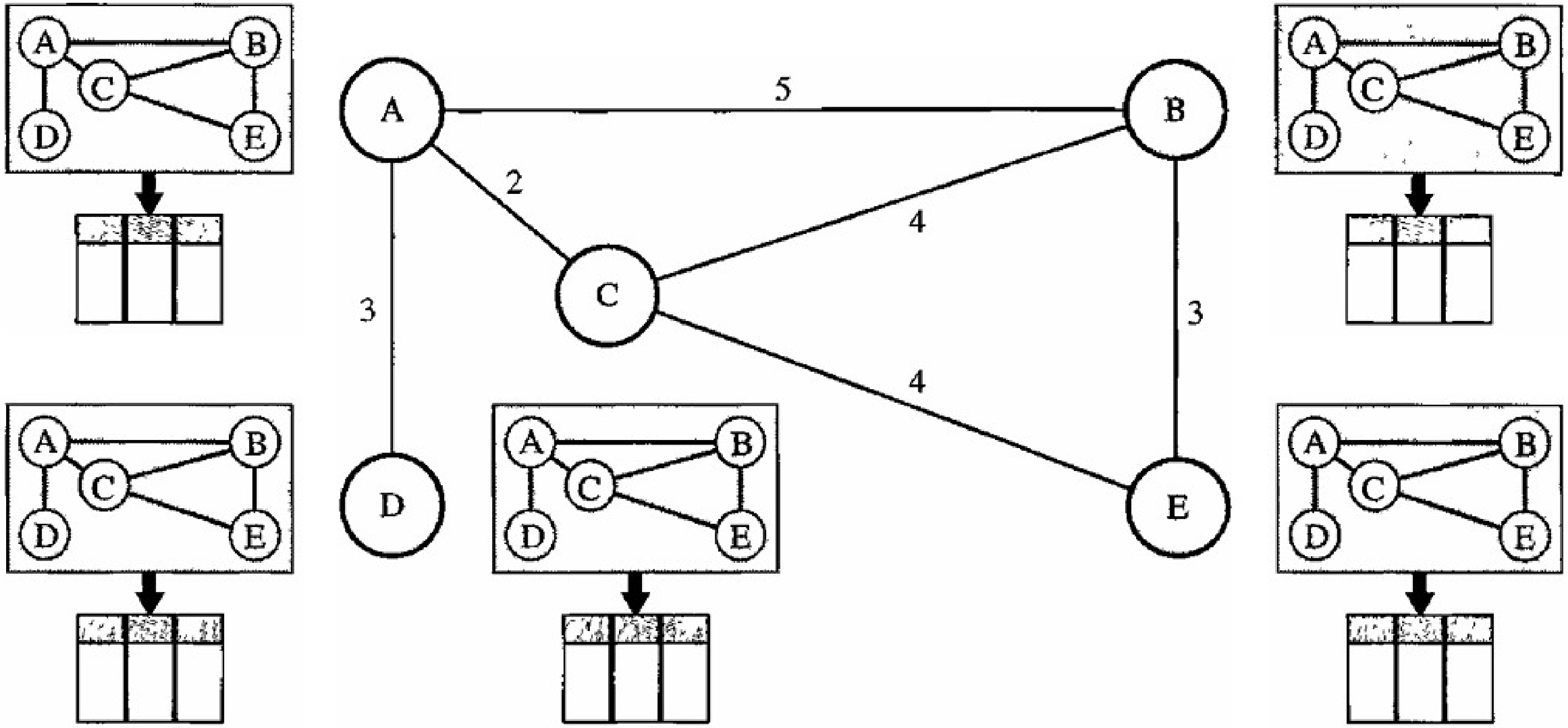


195.2.4.0
130.10.0.0
130.11.0.0
205.5.5.0
195.2.4.1
130.10.0.2
R1
205.5.5.1
R4
R2
130.10.0.1
130.11.0.2
130.11.0.1
195.2.5.1
205.5.6.1
195.2.5.0
195.2.5.2
R3
195.2.6.1
195.2.6.0
205.5.6.0
R2 Table
Dest. | Hop | Next
130.10.0.0 | 1 |
130.11.0.0 | 2 | 130.10.0.2
195.2.4.0 | 1 |
195.2.5.0 | 1 |
195.2.6.0 | 2 | 195.2.5.2
205.5.5.0 | 3 | 130.10.0.2
205.5.6.0 | 3 | 130.10.0.2
R3 Table
Dest. | Hop | Next
130.10.0.0 | 2 | 195.2.5.1
130.11.0.0 | 3 | 195.2.5.1
195.2.4.0 | 2 | 195.2.5.1
195.2.5.0 | 1 |
195.2.6.0 | 1 |
205.5.5.0 | 4 | 195.2.5.1
205.5.6.0 | 4 | 195.2.5.1
RI Table
Des! | Hop | Next
130.10.0.0 | 1 | — —
130.11.0.0 | 1 | — —
195.2.4.0 | 2 | 130.10.0.1
195.2.5.0 | 2 | 130.10.0.1
195.2.6.0 | 3 | 130.10.0.1
205.5.5.0 | 2 | 130.11.0.1
205.5.6.0 | 2 | 130.11.0.1
R4 Table
Dest. | Hop | Next
130.10.0.0 | 2 | 130.11.0.2
130.11.0.0 | 1 |
195.2.4.0 | 3 | 130.11.0.2
195.2.5.0 | 3 | 130.11.0.2
195.2.6.0 | 4 | 130.11.0.2
205.5.5.0 | 1 |
205.5.6.0 | 1 |

Link State Routing

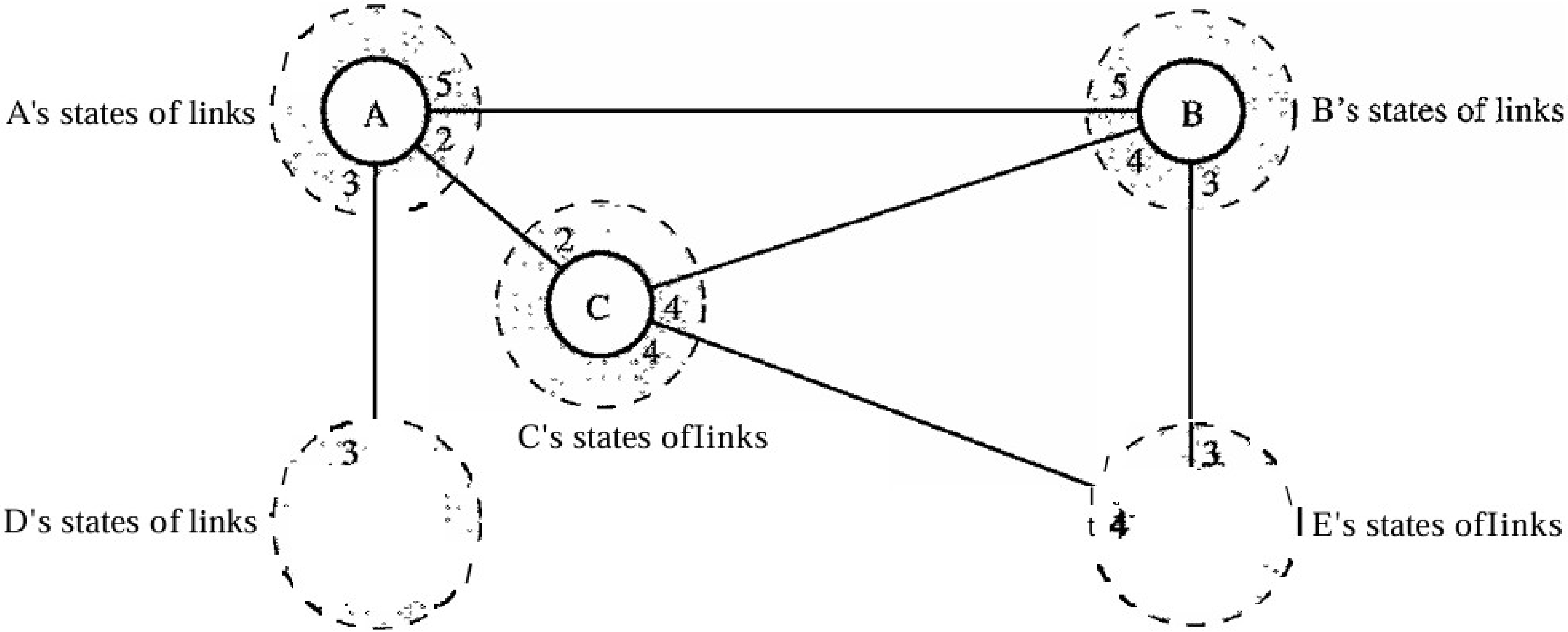
Link state routing has a different philosophy from that of distance vector routing. In link state routing, if each node in the domain has the entire topology of the domain

- the list of nodes and links,
- how they are connected including
- the type of link,
    - cost (metric) of link,
    - condition of the links (up or down)



Link State Routing

Link state routing is based on the assumption that, although the global knowledge about the topology is not clear, each node has partial knowledge and the whole topology can be compiled from the partial knowledge of each node.



Building Routing Tables

In link state routing, four sets of actions are required to ensure that each node has the routing table showing the least-cost node to every other node

1. Creation of the states of the links by each node, called the link state packet (LSP).
2. Dissemination of LSPs to every other router, called flooding, in an efficient and reliable way.
3. Formation of a shortest path tree for each node.
4. Calculation of a routing table based on the shortest path tree

Creation of Link State Packet (LSP)

A link state packet can carry a large amount of information. Assume that it carries following data:

1. The node identity
2. The list of links

The first two are needed to make the topology.

3. A sequence number

Sequence number, facilitates flooding and distinguishes new LSPs from old ones

4. Age.

Age prevents old LSPs from remaining in the domain for a long time.

Creation of Link State Packet (LSP)

LSPs are generated on two occasions

1. When there is a change in the topology of the domain.
2. On a periodic basis.

Creation of Link State Packet (LSP)

Flooding of LSPs

After a node has prepared an LSP, it must be disseminated to all other nodes, not only to its neighbors. The process is called flooding

1. The creating node sends a copy of the LSP out of each interface.
2. A node that receives an LSP compares it with the copy it may already have. If the newly arrived LSP is older than the one it has, it discards the LSP. If it is newer, the node does the following:
   - It discards the old LSP and keeps the new one.
   - It sends a copy of it out of each interface except the one from which the packet arrived. This guarantees that flooding stops somewhere in the domain

Mahindra University
Global Thinkers. Engaged Leaders.

Creation of Link State Packet (LSP)

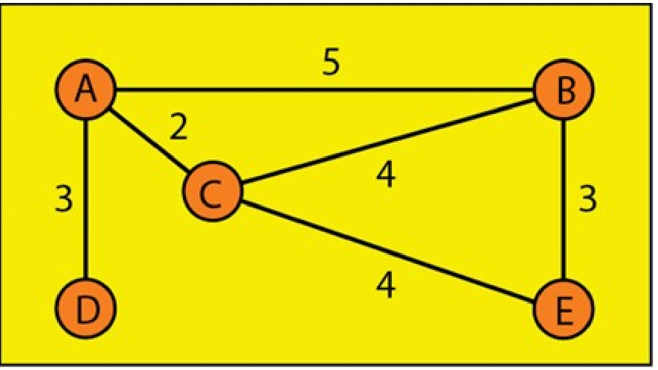
Formation of Shortest Path Tree:

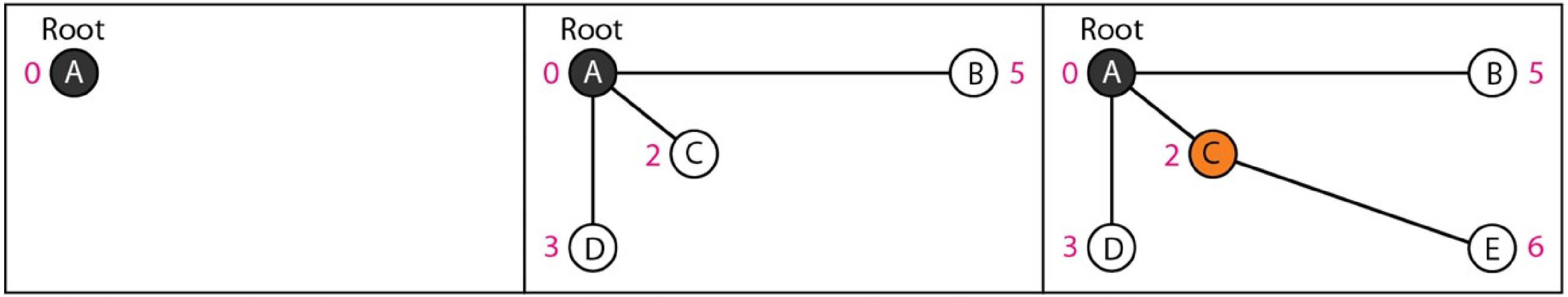
The formation of a shortest path tree using the Dijkstra Algorithm is performed after a node receives all LSPs and learns the complete network topology.

- Set distance of source node = 0, all others = ∞ (infinity).
- Mark all nodes as unvisited; create two sets: tentative and permanent.
- Select the unvisited node with the smallest distance (current node).
- Mark this node as permanent (visited).
- Update distances of all its neighboring nodes:
    - New distance = current node distance + edge cost
    - If new distance is smaller, update it
- Repeat selecting the smallest tentative node and updating neighbors.
- Stop when all nodes are marked permanent or destination is reached.

Creation of Link State Packet (LSP)

Example

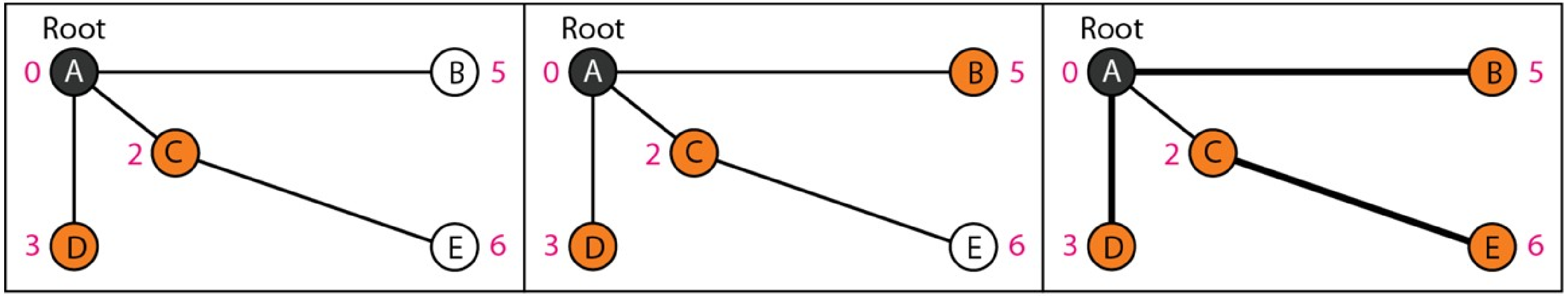




1. Set root to A and move A to tentative list.

2. Move A to permanent list and add B, C, and D to tentative list.

3. Move C to permanent and add E to tentative list.



4. Move D to permanent list.

5. Move B to permanent list.

6. Move E to permanent list (tentative list is empty).

Open Shortest Path First (OSPF)

To handle routing efficiently and in a timely manner, OSPF divides an autonomous system into areas. The area is a collection of

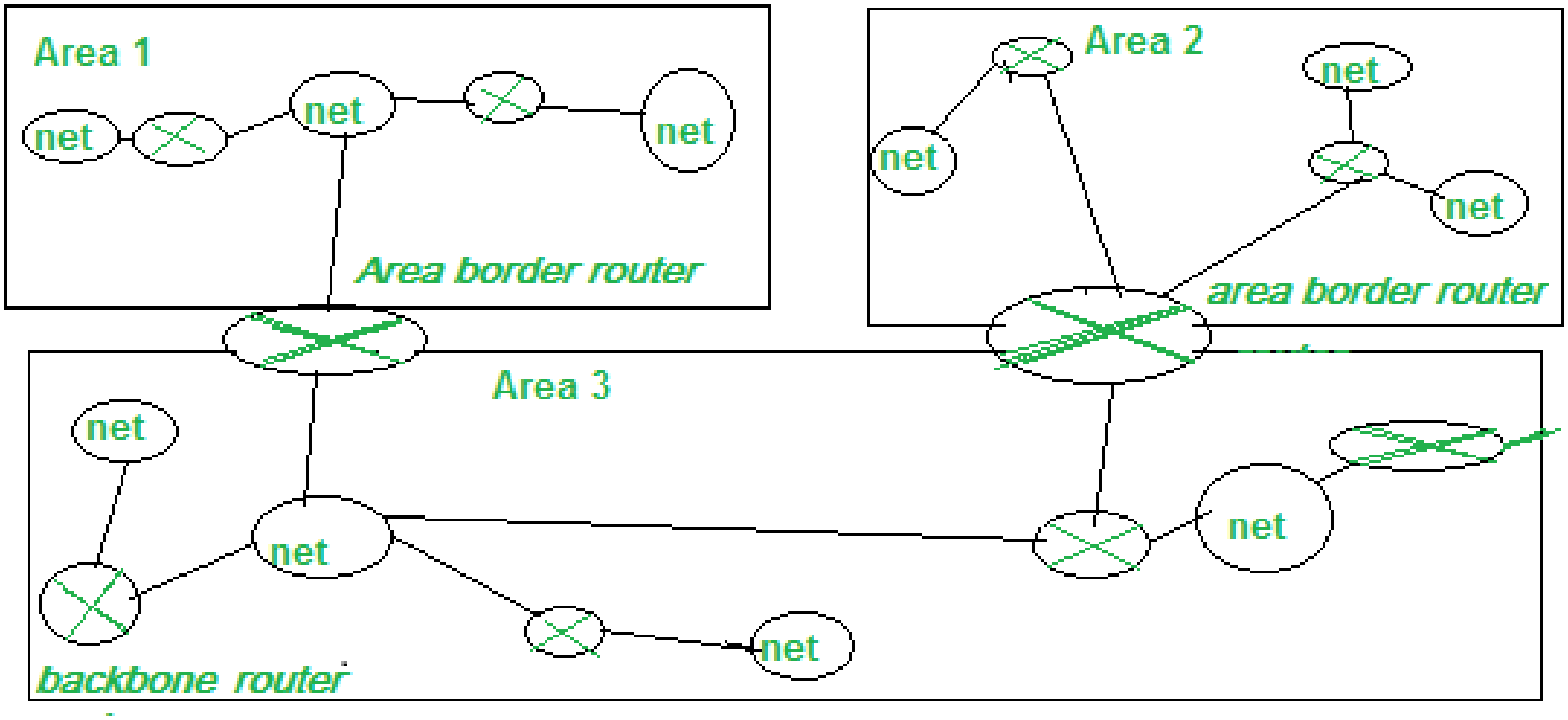
- Networks
- Hosts
- Routers

1. Routers inside an area flood the area with routing information.
2. At the border of an area, special routers called area ***border routers*** summarize the information about the area and send it to other areas.

Among the areas inside an autonomous system is a special area called the *backbone;* all the areas inside an autonomous system must be connected to the backbone.

Open Shortest Path First (OSPF)

Example



Open Shortest Path First (OSPF)

Metric

The OSPF protocol allows the administrator to assign a cost, called the metric, to each route. The metric can be based on a type of service

- Minimum delay,
- Maximum throughput, and
- etc.

Thus a router can have multiple routing tables, each based on a different type of service.

Open Shortest Path First (OSPF)

In OSPF, links represent connections between routers and networks. These are categorized into different types based on how routers are connected.

- *Point-to-Point Link*
  - Direct connection between two routers. No need for DR/BDR election
- *Transit Link (Multi-access Network)*
  - Multiple routers connected to a common network
  - Uses Designated Router (DR) and Backup DR (BDR)
- *Stub Link*
  - A network connected to only one router
  - Used to reach end devices
- *Virtual Link*
  - Logical link between routers through another area
  - Used when an area is not directly connected to backbone (Area 0)